Polynomials (Set B) — MCQ Worksheet

Mathematics • Chapter 2 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. The zero of the linear polynomial $2x + 5$ is:

(A) 5/2 (B) –5/2
(C) 2/5 (D) –2/5

Q2. If $p(x) = x^2 - 3x + 2$, then $p(1) =$

(A) 0 (B) 1
(C) 2 (D) –2

Q3. If α and β are zeros of $x^2 - 5x + 6$, then $\alpha + \beta =$

(A) 5 (B) –5
(C) 6 (D) –6

Q4. If α and β are zeros of $x^2 - 5x + 6$, then $\alpha\beta =$

(A) 5 (B) –5
(C) 6 (D) –6

Q5. The degree of the polynomial $7 - x + 2x^2$ is:

(A) 0 (B) 1
(C) 2 (D) 3

Q6. A quadratic polynomial with sum of zeros 4 and product 3 is:

(A) $x^2 - 4x + 3$ (B) $x^2 + 4x + 3$
(C) $x^2 - 4x - 3$ (D) $x^2 - 3x + 4$

Q7. Zeros of $x^2 - 16$ are:

(A) 4 and –4 (B) 16 and –16
(C) 4 only (D) 0 and 16

Q8. If one zero of $x^2 + 5x + k$ is –3, then $k =$

(A) –6 (B) 6
(C) 3 (D) –3

Q9. The number of zeros of polynomial $y = x^4$ is:

(A) 1 (B) 2
(C) 3 (D) 4

Q10. If $p(x) = 2x^3 - 3x^2 + 5$, then $p(0) =$

(A) 0 (B) 2
(C) 5 (D) –3

Q11. If α and β are zeros of $x^2 + 7x + 10$, then $\alpha^2 + \beta^2 =$

(A) 29 (B) 39
(C) 49 (D) 59

Q12. The zeros of polynomial $2x^2 + x - 3$ are:

(A) 1 and –3/2 (B) –1 and 3/2
(C) 2 and –3 (D) –2 and 3

Q13. The graph of $y = ax^2 + bx + c$ is a parabola opening upward when:

(A) $a > 0$ (B) $a < 0$
(C) $a = 0$ (D) $b > 0$

Q14. If 2 and 3 are zeros of polynomial $2x^3 - 13x^2 + bx - 6$, then $b =$

(A) 22 (B) 23
(C) 26 (D) 27

Q15. If α, β are zeros of polynomial $p(x) = x^2 - 6x + 4$, then $\alpha/\beta + \beta/\alpha =$

(A) 7 (B) 11
(C) 13 (D) 14

Q16. If α, β are zeros of $x^2 - p(x + 1) - c$, then $(\alpha + 1)(\beta + 1) =$

(A) $c - 1$ (B) $1 - c$
(C) c (D) $1 + c$

Q17. If α and β are zeros of polynomial $6y^2 - 7y + 2$, the polynomial whose zeros are $1/\alpha$ and $1/\beta$ is:

(A) $2y^2 - 7y + 6$ (B) $6y^2 + 7y + 2$
(C) $2y^2 + 7y + 6$ (D) $y^2 - 7y + 6$

Q18. Find a quadratic polynomial whose zeros are $2 + \sqrt{3}$ and $2 - \sqrt{3}$.

(A) $x^2 - 4x + 1$ (B) $x^2 + 4x - 1$
(C) $x^2 - 4x - 1$ (D) $x^2 + 4x + 1$

Q19. On dividing $x^3 - 3x^2 + x + 2$ by polynomial $g(x)$, the quotient is $x - 2$ and remainder is $-2x + 4$. Find $g(x)$.

(A) $x^2 - x + 1$ (B) $x^2 + x + 1$
(C) $x^2 - x - 1$ (D) $x^2 + x - 1$

Q20. If α, β, γ are zeros of $x^3 - 6x^2 + 11x - 6$, then $\alpha\beta + \beta\gamma + \gamma\alpha =$

(A) 6 (B) 11
(C) –11 (D) –6

Q21. If both 2 and 0 are zeros of $px^2 + 5x + 4$, find p.

(A) Cannot exist ($4 \neq 0$) (B) $p = -5/2$
(C) $p = 0$ (D) $p = 1$

Q22. If α, β are zeros of polynomial $x^2 + 4x + 3$, the polynomial whose zeros are $1 + \beta/\alpha$ and $1 + \alpha/\beta$ is:

(A) $3x^2 - 16x + 16$ (B) $3x^2 + 16x + 16$
(C) $x^2 + 16x + 3$ (D) $3x^2 + 4x + 16$

Q23. If polynomial $6x^4 + 8x^3 + 17x^2 + 21x + 7$ is divided by $3x^2 + 4x + 1$, remainder is $ax + b$. Then $a + b =$

(A) 3 (B) 4
(C) 5 (D) 7

Q24. If α, β are zeros of $x^2 - x - 6$, find quadratic polynomial whose zeros are $2\alpha + 1$ and $2\beta + 1$.

(A) $x^2 - 4x - 21$ (B) $x^2 + 4x - 21$
(C) $x^2 - 4x + 21$ (D) $x^2 - x - 21$

Q25. Find a polynomial whose zeros are $-2/\sqrt{3}$ and $\sqrt{3}/4$.

(A) $12x^2 + 5x - 6$ (B) $4\sqrt{3}\,x^2 + 5x - 2$
(C) $12x^2 - 5x - 6$ (D) $4\sqrt{3}\,x^2 - 5x - 2$

— End of Worksheet —